M'Hamed Bougara University of Boumerdes **2026-2027**

Faculty of Sciences – Computer Science Department

Promotion : M1 – LMD. **Module : Apprentissage automatique**

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Lab 1: Introduction to Python and Anaconda

1- Objectives of the lab

1- Set up Python using Anaconda, and how to use Jupyter Notebooks.
2- Execute basic Python instructions and basic libraries such as NumPy and Pandas.
3- Load and visualize data.

Basic Python Instructions Execute the following instructions and observe the results.

```
# Variables
x = 15
y = 3.25
name = "FAI Lab"
print(x, y, name)
# Conditional statement
if x > 5:
    print(f"{x} is greater than 5")
else:
    print(f"{x} is less than or equal to 5")
# Loop
for i in range(5):
   print(i)
# Functions:
def square(nbr):
   return nbr * nbr
print(square(4))
```

Using Basic Libraries

Numpy : (Numerical Python) is a powerful library in Python primarily used for numerical computing. It provides essentially large, multi-dimensional arrays and matrices, along with a collection of mathematical functions to operate on these arrays. It offers efficient storage and operations on large datasets, and facilitates a variety of mathematical and statistical operations. As an example, execute the following instructions and observe the results.

```
import numpy as np
# Create a NumPy array
array = np.array([1, 2, 3, 4, 5])
print(array)
```

```
# Basic operations with NumPy arrays
print("Sum:", np.sum(array))
print("Mean:", np.mean(array))
```

Pandas : Pandas is a Python library used for data manipulation and analysis by providing mainly two data structures: Series and Dagtaframes. A DataFrame is two-dimensional labeled data structure (like a table with rows and columns). Pandas helps manipulating datasets, data cleaning and analysis.

```
import pandas as pd

# Create a simple DataFrame

data = {'Name': ['Ali', 'Karima', 'Omar'],

    'Age': [24, 27, 22],

    'Score': [88, 92, 95]}
```

```
df = pd.DataFrame(data)

print(df)

# Access a column

print(df['Name'])
```

Loading and Visualizing Data

matplotlib : a Python library used for creating static, animated, and interactive visualizations. It is widely used to generate plots, charts, and graphs , histograms...

```
dflab1 = pd.read_csv('Salary_Data.csv')

print(dflab1.head())

import matplotlib.pyplot as plt

# Simple line plot with Matplotlib

plt.plot(dflab1['YearsExperience'], dflab1['Salary'], 'o-')

plt.xlabel('Experience')

plt.ylabel('Salary')

plt.title('Experience vs Salary')

plt.show()

# Histogram with Matplotlib

plt.hist(dflab1['YearsExperience'], bins=10, alpha=0.7, color='blue')

plt.xlabel('Experience')

plt.ylabel('Frequency')

plt.title('Experience distribution')

plt.show()
```